



Capstone Project 6

Title: Agentic Supply Chain Disruption Predictor & Simulation Engine

Overview and Problem Statement:

This project aims to develop an AI-powered supply chain disruption prediction and simulation system that proactively monitors global risk signals, predicts potential supply chain disruptions, forecasts demand impact, and simulates mitigation scenarios using multi-agent AI workflows, Retrieval-Augmented Generation (RAG), time-series forecasting, and discrete-event simulation techniques.

Modern supply chains are highly interconnected networks involving suppliers, manufacturers, logistics providers, ports, warehouses, and retailers across multiple countries. Disruptions such as extreme weather events, geopolitical conflicts, labor strikes, port closures, tariffs, or factory shutdowns can significantly affect inventory availability, delivery timelines, and revenue.

Most organizations react only after disruptions occur because monitoring large-scale real-time data sources manually is difficult and inefficient. Businesses require intelligent systems capable of continuously analyzing news feeds, weather alerts, shipping indices, and logistics signals to identify risks before they escalate.

The proposed system will build a LangGraph-orchestrated multi-agent platform that continuously ingests disruption-related information, classifies supply chain risks, forecasts demand fluctuations, simulates disruption scenarios, and generates mitigation recommendations. The platform aims to support proactive risk management, improve operational resilience, and assist organizations in strategic supply chain planning.

Dataset:

The project will utilize a combination of public datasets, APIs, and synthetic disruption datasets:

1. SupplyChainNet Dataset:
 - a. Source: Kaggle Supply Chain datasets
 - b. Content: Historical supply chain transactions, shipping records, supplier information, logistics delays, and disruption events
[Supply Chain Dataset \(Kaggle\)](#)
2. Freightos Baltic Index Data:
 - a. Source: Public Freightos shipping index



- b. Content: Container shipping rate indices, freight trends, and logistics cost fluctuations
[Freightos Baltic Index](#)
- 3. Open-Meteo Weather API:
 - a. Source: Open-Meteo API
 - b. Content: Historical and forecast weather data for logistics hubs and shipping regions
[Open-Meteo API](#)
- 4. News and RSS Feed Data:
 - a. Source: Reuters, Bloomberg, Supply Chain Dive, and public RSS feeds
 - b. Content: News articles related to labor strikes, geopolitical risks, tariffs, factory shutdowns, and logistics disruptions
[Reuters Business News RSS](#), [Google News API](#)
- 5. Synthetic Supply Chain Events:
 - a. Source: AI-generated synthetic disruption events
 - b. Content: Simulated demand shocks, disruption narratives, supplier failures, and logistics incidents

Methodology:

- 1. Real-Time Data Ingestion Agent:
 - a. Continuously collect data from RSS feeds, logistics APIs, weather APIs, and shipping indices
 - b. Extract disruption-related events using feedparser and NLP pipelines
 - c. Store incoming disruption signals in structured format
- 2. News and Event Analysis Agent:
 - a. Analyze news articles and logistics alerts using LLMs
 - b. Identify disruption categories such as weather, geopolitical, logistics, raw material shortages, and demand shocks
- 3. Weather Risk Monitoring Agent:
 - a. Fetch weather forecasts using Open-Meteo API
 - b. Detect extreme weather events affecting ports, transportation hubs, and manufacturing facilities
- 4. Risk Classification Agent:
 - a. Build LangGraph-based orchestration workflow
 - b. Use DistilBERT-based classifiers to categorize supply chain risks
 - c. Generate supplier-level and trade-lane risk scores
- 5. Demand Forecasting Agent:
 - a. Train forecasting models using Facebook Prophet
 - b. Incorporate disruption risk signals into demand forecasting pipelines



- c. Predict demand deviations and inventory fluctuations
- 6. Simulation Agent:
 - a. Simulate supply chain disruptions using SimPy discrete-event simulation
 - b. Model suppliers, warehouses, ports, and retailers as network nodes
 - c. Run Monte Carlo simulations to estimate stockout probability and revenue impact
- 7. Mitigation Recommendation Agent:
 - a. Generate natural-language mitigation recommendations using LLMs
 - b. Suggest alternate suppliers, route changes, safety stock adjustments, and inventory actions
- 8. Dashboard and Alerting:
 - a. Build Gradio dashboard for risk visualization and scenario analysis
 - b. Display disruption heatmaps, timelines, and simulation outcomes
 - c. Trigger alerts for high-risk scenarios
- 9. Evaluation:
 - a. Measure risk classification accuracy
 - b. Evaluate demand forecast deviation
 - c. Assess simulation realism and mitigation recommendation quality

Challenges:

1. Real-Time Data Integration: Continuously process multiple external data sources reliably
2. Risk Signal Extraction: Identify meaningful disruption indicators from noisy news and weather data
3. Forecasting Uncertainty: Handle uncertainty in demand forecasting under disruption conditions
4. Simulation Complexity: Model realistic supply chain behavior and interconnected dependencies
5. Multi-Agent Coordination: Synchronize ingestion, forecasting, simulation, and mitigation agents effectively
6. Scalability: Efficiently manage large-scale supplier networks and logistics data
7. Recommendation Reliability: Generate actionable and business-relevant mitigation strategies

References:

1. Facebook Prophet Documentation <https://facebook.github.io/prophet/>
2. SimPy Documentation <https://simpy.readthedocs.io/en/latest/>
3. Feedparser Documentation <https://feedparser.readthedocs.io/en/latest/>
4. DistilBERT Model Documentation <https://huggingface.co/distilbert-base-uncased>